7.3: Passive transport is diffusion of a substance across a membrane with no energy investment

Molecules have a type of energy called thermal energy, due to their constant motion (see Concept 3.2). One result of this motion is **diffusion**, the movement of particles of any substance so that they spread out into the available space. Each molecule moves randomly, yet diffusion of a *population* of molecules may be directional. Imagine a synthetic membrane separating pure water from a solution of a dye in water. Study **Figure 7.11a** carefully to see how diffusion would result in equal concentrations of dye molecules in both solutions. At that point, a dynamic equilibrium will exist, with as many dye molecules crossing per second in one direction as in the other.

Figure 7.11

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Diffusion of solutes across a synthetic membrane.
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Diffusion of solutes across a synthetic membrane.

Each large arrow shows net diffusion of dye molecules of that color.

Parts are as follows.

Part a. Diffusion of one solute. The membrane has pores large enough for molecules of dye to pass through. Random movement of dye molecules will cause some to pass through the pores; this will happen more often on the side with more dye molecules. The dye diffuses from where it is more concentrated to where it is less concentrated (called diffusing down a concentration gradient). This leads to a dynamic equilibrium: The solute molecules continue to cross the membrane, but at roughly equal rates in both directions. In the first image, there is a high concentration of dye molecules on one side of a synthetic membrane and no dye molecules on the other side. Small arrows within the image indicate that the dye molecules are in motion, and an arrow beneath the image indicates that the net direction for the diffusion of the dye molecules is from high concentration to low (no) concentration across the membrane. In the second image, some dye molecules have diffused across the membrane, but the molecules are still more concentrated on one side. Arrows indicate that the direction of net diffusion will continue as dye molecules move from high to low concentration across the membrane. In the final image the concentration of dye molecules is equal on both

sides of the membrane. Arrows indicate that the dye molecules are now at equilibrium, meaning they will still show motion, but there will be no more net movement of dye molecules across the membrane.

Part b. Diffusion of two solutes. Solutions of two different dyes are separated by a membrane that is permeable to both. Each dye diffuses down its own concentration gradient. There will be a net diffusion of the purple dye toward the left, even though the total solute concentration was initially greater on the left side. In the first image, one molecule is concentrated on one side of a membrane, while a different molecule is concentrated on the other side. Arrows indicate that both molecules are in motion, and that each will begin to diffuse down its own concentration gradient. In the second image, diffusion progresses independently for each molecule. Both molecules diffuse across the membrane down their own concentration gradients regardless of the concentration of the other molecule. In the third image, equilibrium for both molecules has been reached, with both equally concentrated on both sides of the membrane.

Animation

Diffusion

image

image

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Here is a simple rule of diffusion: In the absence of any other forces, a substance will diffuse from where it is more concentrated to where it is less concentrated. Put another way, a substance diffuses down its **concentration gradient**, the region along which the density of a chemical substance increases or decreases (in this case, decreases). Diffusion is a spontaneous process, needing no input of energy. Each substance diffuses down its *own* concentration gradient, unaffected by the concentration gradients of other substances **(Figure 7.11b)**.

Much of the traffic across cell membranes occurs by diffusion. When a substance is more concentrated on one side of a membrane than on the other, there is a tendency for it to diffuse across, down its concentration gradient (assuming that the membrane is permeable to that substance). One important example is the uptake of oxygen by a cell performing cellular respiration. Dissolved oxygen diffuses into the cell across the plasma membrane. As long as cellular respiration consumes the O2 as it enters, diffusion into the cell will continue because the concentration gradient favors movement in that direction.

The diffusion of a substance across a biological membrane is called **passive transport** because it requires no energy. The concentration gradient itself represents potential energy (see Concept 2.2 and Figure 8.5b) and drives diffusion. Remember, though, that membranes are selectively permeable and therefore have

different effects on the rates of diffusion of various molecules. Water can diffuse very rapidly across the membranes of cells with aquaporins, compared with diffusion in the absence of aquaporins. The movement of water across the plasma membrane has important consequences for cells.

Effects of Osmosis on Water Balance

To see how two solutions with different solute concentrations interact, picture a U-shaped glass tube with a selectively permeable artificial membrane separating two sugar solutions **(Figure 7.12)**. Pores in this synthetic membrane are too small for sugar molecules to pass through but large enough for water molecules. However, tight clustering of water molecules around the hydrophilic solute molecules makes some of the water unavailable to cross the membrane. As a result, the solution with a higher solute concentration has a lower *free* water concentration. Water diffuses across the membrane from the region of higher free water concentration (lower solute concentration) to that of lower free water concentration (higher solute concentration) until the solute concentrations on both sides of the membrane are more nearly equal. The diffusion of free water across a selectively permeable membrane, whether artificial or cellular, is called **osmosis**. The movement of water across cell membranes and the balance of water between the cell and its environment are crucial to organisms. Let's now apply what we've learned about osmosis in this system to living cells.

Figure 7.12

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Osmosis.
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An illustration of osmosis.

Two sugar solutions of different concentrations are separated by a membrane that the solvent (water) can pass through but the solute (sugar) cannot. Water molecules move randomly and may cross in either direction, but overall, water diffuses from the solution with less concentrated solute to that with more concentrated solute. This passive transport of water, or osmosis, makes the sugar concentrations on both sides roughly equal.

Two U shaped tubes filled with solutions of different sugar concentrations separated by a selectively permeable membrane at the base of the U tube are depicted. The first U shaped tube shows the initial starting point of the experiment where fluid levels on either side of the membrane are equal. Arrows indicate that water will move by osmosis across the membrane from the solution with lower solute (sugar) concentration to the solution with higher solute concentration. The result, shown in a second image depicting the U shaped tube, is that as water moves across the membrane from low to high solute

concentration, the fluid level rises with the influx of water, and is higher than the fluid level on the other side of the membrane. After osmosis, although the fluid level is higher on the side of the membrane that originally had the higher solute concentration, solute concentrations are similar on both sides of the membrane. A third image is depicted by zooming in on the membrane area of the U shaped tube and gives a closer look at water and solute molecules during osmosis on both sides of the membrane. On the side of the membrane with a lower solute concentration, there are fewer solute molecules and more free water molecules (water molecules not interacting with solute molecules). From this side, water molecules can pass through pores, but sugar molecules cannot. On the side of the membrane with a higher solute concentration, there are more solute molecules and fewer free water molecules, as more water molecules are clustered around the sugar molecules rather than moving freely. Thus, osmosis proceeds from lower solute concentration to higher solute concentration, as there are more free water molecules in the solution with lower solute concentration that can move across the membrane. Water moves from an area of higher to lower free water concentration (lower to higher solute concentration).

VISUAL SKILLS *If an orange dye capable of passing through the membrane was added to the left side of the tube above, how would it be distributed at the end of the experiment? (See*Figure 7.11*.)* Would the final solution levels in the tube be affected? What cellular component does the membrane represent in this experiment?*

*For suggested answer, see*Appendix A*.*

Figure 7.12 Walkthrough

Osmosis

Watch an explanation of this figure in the Figure 7.12 Walkthrough: Osmosis.

image

Play

image

Water Balance of Cells Without Cell Walls

To explain the behavior of a cell in a solution, we must consider both solute concentration and membrane permeability. Both factors are taken into account in the concept of **tonicity**, the ability of a surrounding solution to cause a cell to gain or lose water. The tonicity of a solution depends in part on its concentration of solutes that cannot cross the membrane (nonpenetrating solutes) relative to that inside the cell. If there is a higher concentration of nonpenetrating solutes in the surrounding solution, water will tend to leave the cell, and vice versa.

If a cell without a cell wall, such as an animal cell, is immersed in an environment that is **isotonic** to the cell (*iso* means “same”), there will be no *net* movement of water across the plasma membrane. Water diffuses across the membrane, but at the same rate in both directions. In an isotonic environment, the volume of an animal cell is stable **(Figure 7.13a)**.

Figure 7.13

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The water balance of living cells.
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The water balance of living animal and plant cells.

How living cells react to changes in the solute concentration of their environment depends on whether or not they have cell walls. ***(a)*** *Animal cells, such as this red blood cell, do not have cell walls.* ***(b)*** *Plant cells do have cell walls. (Arrows indicate net water movement after the cells were first placed in these solutions.)*

Part a. Animal cell. An animal cell fares best in an isotonic environment unless it has special adaptations that offset the osmotic uptake or loss of water. Water flows into animal cells in hypotonic solutions, and may result is cells lysing (bursting). There is no net flow of water in or out of the cell in isotonic solutions; this condition is normal for animal cells. In hypertonic solutions, water leaves the animal cell and the animal cell shrivels.

Part b. Plant cell. Plant cells are turgid (firm) and generally healthiest in a hypotonic environment, where the uptake of water is eventually balanced by the wall pushing back on the cell. Water flows into plant cells in hypotonic solutions, creating turgid (normal) conditions for plant cells as water pushes against the cell wall. In isotonic solutions, there is no net flow of water in or out of the cell and the plant cell is flaccid. In hypertonic solutions, water flows out of the cell and the plant cell can be plasmolyzed.

Why do limp celery stalks become crisp when placed in a glass of water?

*For suggested answer, see*Appendix A.**

Animation

Osmosis and Water Balance in Cells

image

Play

image

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Video

Turgid*Elodea*

image

image

Video

Plasmolysis in *Elodea*

image

image

Let's transfer the cell to a solution that is **hypertonic** to the cell (*hyper* means "more," in this case referring to nonpenetrating solutes). The cell will lose water, shrivel, and probably die. This is why an increase in the salinity (saltiness) of a lake can kill the animals there; if the lake water becomes hypertonic to the animals' cells, they might shrivel and die. However, taking up too much water can be just as hazardous to a cell as losing water. If we place the cell in a solution that is **hypotonic** to the cell (*hypo* means "less"), water will enter the cell faster than it leaves, and the cell will swell and lyse (burst) like an overfilled water balloon.

A cell without rigid cell walls can't tolerate either excessive uptake or excessive loss of water. This problem of water balance is automatically solved if such a cell lives in isotonic surroundings. Seawater is isotonic to many marine invertebrates. The cells of most terrestrial (land-dwelling) animals are bathed in an extracellular fluid that is isotonic to the cells. In hypertonic or hypotonic environments, however, organisms that lack rigid cell walls must have other adaptations for **osmoregulation**, the control of solute concentrations and water balance. For example, the unicellular protist *Paramecium caudatum* lives in pond water, which is hypotonic to the cell. *Paramecium* has a plasma membrane that is much less permeable to water than the membranes of most other cells, but this only slows the uptake of water, which continually enters the cell. The reason the *Paramecium* cell doesn't burst is that it has a contractile vacuole, an organelle that functions as a pump to force water out of the cell as fast as it enters by osmosis **(Figure 7.14)**. In contrast, the bacteria and archaea that live in hypersaline (excessively salty) environments (see Figure 27.1) have cellular mechanisms that balance the internal and external solute concentrations to ensure that water does not move out of the cell. We'll examine other evolutionary adaptations for osmoregulation by animals in Concept 44.1.

Figure 7.14

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The contractile vacuole of *Paramecium*.
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A Paramecium with contractile vacuoles is shown on a 50 micrometer scale. It is approximately 200 micrometers in length.

The vacuole collects fluid from canals in the cytoplasm. When full, the vacuole and canals contract, expelling fluid from the cell (LM).

Video

Paramecium Vacuole

image

image

Water Balance of Cells with Cell Walls

The cells of plants, prokaryotes, fungi, and some protists are surrounded by cell walls (see Figure 6.27). When such a cell is immersed in a hypotonic solution—bathed in rainwater, for example—the cell wall helps maintain the cell's water balance. Consider a plant cell. Like an animal cell, the plant cell swells as water enters by osmosis **(Figure 7.13b)**. However, the relatively inelastic cell wall will expand only so much before it exerts a back pressure on the cell, called *turgor pressure*, that opposes further water uptake. At this point, the cell is **turgid** (very firm), which is the healthy state for most plant cells. Plants that are not woody, such as most houseplants, depend for mechanical support on cells kept turgid by a surrounding hypotonic solution. If a plant's cells and surroundings are isotonic, there is no net tendency for water to enter and the cells become **flaccid** (limp); the plant wilts.

However, a cell wall is of no advantage if the cell is immersed in a hypertonic environment. In this case, a plant cell, like an animal cell, will lose water to its surroundings and shrink. As the plant cell shrivels, its plasma membrane pulls away from the cell wall at multiple places. This phenomenon, called **plasmolysis**, causes the plant to wilt and can lead to plant death. The walled cells of bacteria and fungi also plasmolyze in hypertonic environments.

Facilitated Diffusion: Passive Transport Aided by Proteins

Let's look more closely at how water and certain hydrophilic solutes cross a membrane. As mentioned earlier, many polar molecules and ions blocked by the lipid bilayer of the membrane diffuse passively with the help of transport proteins that span the membrane. This phenomenon is called **facilitated diffusion**.

Cell biologists are still trying to learn exactly how various transport proteins facilitate diffusion. Most transport proteins are very specific: They transport some substances but not others.

As mentioned earlier, the two types of transport proteins are channel proteins and carrier proteins. Channel proteins simply provide corridors that allow specific molecules or ions to cross the membrane **(Figure 7.15a)**. The hydrophilic passageways provided by these proteins can allow water molecules or small ions to diffuse very quickly from one side of the membrane to the other. Aquaporins, the water channel proteins, facilitate the massive levels of diffusion of water (osmosis) that occur in plant cells and in animal cells such as red blood cells (see Figure 7.13). Certain kidney cells also have a high number of aquaporins, allowing them to reclaim water from urine before it is excreted. If the kidneys didn't perform this function, you would excrete about 180 L of urine per day—and have to drink an equal volume of water!

Figure 7.15

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    Two types of transport proteins that carry out facilitated
diffusion.
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Diagrams of a channel protein and carrier protein.

In both cases, the protein can transport the solute in either direction, but the net movement is down the concentration gradient of the solute.

Part a. A channel protein has a channel through which water molecules or a specific solute can pass. Water molecules move across the membrane from higher concentration in the extracellular fluid to lower concentration in the cytoplasm through a hydrophilic channel in a channel protein.

Part b. A carrier protein alternates between two shapes, moving a solute across the membrane during the shape change. Solutes move from higher concentration in the extracellular fluid to lower concentration in the cytoplasm by moving through a carrier protein in the membrane, which changes shape to transport them.

Animation

Facilitated Diffusion

image

image

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Channel proteins that transport ions are called **ion channels**. Many ion channels function as **gated channels**, which open or close in response to a stimulus (see Figure 11.8). For some gated channels, the stimulus is electrical. In a nerve cell, for example, a potassium ion channel protein (see computer model) opens in response to an electrical stimulus, allowing a stream of potassium ions to leave the cell. This restores the cell's ability to fire again. Other gated channels have a chemical stimulus: They open or close when a specific substance (not the one to be transported) binds to the channel. Ion channels are important in the functioning of the nervous system, as you'll learn in Chapter 48.

A potassium ion is at the center of a protein.

Potassium ion channel protein (See also the side view of a calcium channel protein in Figure 6.32a.)

Interview

Interview with Elba Serrano: Investigating how ion channels enable you to hear

A photo of Elba Serrano.

Carrier proteins, such as the glucose transporter mentioned earlier, seem to undergo a subtle change in shape that somehow translocates the solute-binding site across the membrane **(Figure 7.15b)**. Such a change in shape may be triggered by the binding and release of the transported molecule. Like ion channels, carrier proteins involved in facilitated diffusion result in the net movement of a substance down its concentration gradient. No energy input is thus required: This is passive transport. The **Scientific Skills Exercise** gives you an opportunity to work with data from an experiment related to glucose transport.

BioFlix® Animation

Passive Transport

image

Scientific Skills Exercise

Interpreting a Scatter Plot with Two Sets of Data

AP Science Practices 2, 3, 4, 6

Is Glucose Uptake into Cells Affected by Age? Glucose, an important energy source for animals, is transported into cells by facilitated diffusion using protein carriers. In this exercise, you will interpret a

graph with two sets of data from an experiment that examined glucose uptake over time in red blood cells from guinea pigs of different ages. You will determine if the cells' rate of glucose uptake depended on the age of the guinea pig.

How the Experiment Was Done Researchers incubated guinea pig red blood cells in a 300 mM (millimolar) radioactive glucose solution at pH 7.4 at 25°C. Every 10 or 15 minutes, they removed a sample of cells and measured the concentration of radioactive glucose inside those cells. The cells came from either a 15-day-old or a 1-month-old guinea pig.

Data from the Experiment When you have multiple sets of data, it can be useful to plot them on the same graph for comparison. In the graph here, each set of dots (of the same color) forms a *scatter plot*, in which every data point represents two numerical values, one for each variable. For each data set, a curve that best fits the points has been drawn to make it easier to see the trends. (For additional information about graphs, see the Scientific Skills Review in Appendix D.)

Two graphs plot concentration of radioactive glucose versus Incubation time in minutes.

Glucose Uptake over Time in Guinea Pig Red Blood Cells.

Data from T. Kondo and E. Beutler, Developmental changes in glucose transport of guinea pig erythrocytes, *Journal of Clinical Investigation* 65:1–4 (1980).

The graph of a 15-day-old guinea pig rises from (0, 0) through (10, 50), (20, 72), (30, 85), and (50, 97) to (60, 102). The graph for a 1-month-old-guinea pig rises from (0, 0) through (10, 20), (20, 35), (30, 40), and (45, 48) to (60, 55). All values are estimated.

INTERPRET THE DATA

- First make sure you understand the parts of the graph.
- Which variable is the independent variable—the variable controlled by the researchers?
- Which variable is the dependent variable—the variable that depended on the treatment and was measured by the researchers?
- What do the red dots represent?
- The blue dots?
- From the data points on the graph, construct a table of the data. Put "Incubation Time (min)" in the left column of the table.
- What does the graph show? Compare and contrast glucose uptake in red blood cells from 15-day-old and 1-month-old guinea pigs.

- Develop a hypothesis to explain the difference between glucose uptake in red blood cells from 15-day-old and 1-month-old guinea pigs. (Think about how glucose gets into cells.)
- Design an experiment to test your hypothesis.

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.